NNAMDI AZIKIWE UNIVERSITY

FACULTY OF BIOSCIENCES

BIO 102 - INTRODUCTORY BIOLOGY II

INTODUCTION

Living things stay alive by doing certain activities that keep them going. These activities help them control what they do every day. One of these activities is called excretion. Inside the cells of living things, a process called metabolism happens. This process produces both helpful and harmful substances. If harmful substances build up, they can hurt the living thing. So, living things get rid of these waste products from their bodies, and this is called excretion. Different living things get rid of waste in different ways. This lecture will look at how plants get rid of waste and how it's different from animals.

Chemical reactions occur in the cells of living organisms all the time to carry out the life processes. The sum of these reactions is called metabolism. Metabolism produces useful products as well as toxic (poisonous) by-products. These toxic substances have to be removed as they are harmful if allowed to accumulate. The removal of metabolic waste products from the body of an organism is known as excretion.

The major excretory products are carbon dioxide, excess water, and nitrogenous compounds like ammonia, urea, uric acid, etc. Carbon dioxide and water are produced in the process of tissue respiration. Nitrogenous compounds are formed from the breakdown of proteins and amino acids. Water and salts in excess of the body's needs are also excreted.

We acquire most of the water with our food and drink and some by metabolism, e.g., the water produced during cellular respiration. Other excretory products include chemicals from medicines, toxic substances, and circulating hormones that have already served their purpose. We will learn how metabolic wastes get eliminated.

Excretion in animals

Many unicellular organisms like Amoeba throw out their wastes by diffusion from their body surface. Protozoan's have no organs for excretion. As they live in an aquatic habitat, their wastes are eliminated by diffusion through the plasma membrane.

Simple multicellular organisms like Hydra throw out solid waste matter through their mouth. Higher multicellular organisms have well-defined specialized excretory organs. These organs could be simple tubular structures as in flatworms and leech.

The excretory organs of insects (e.g., grasshopper, cockroach and housefly) are also tubular. They remove nitrogenous wastes from the body fluid and help in maintaining the water balance in the body.

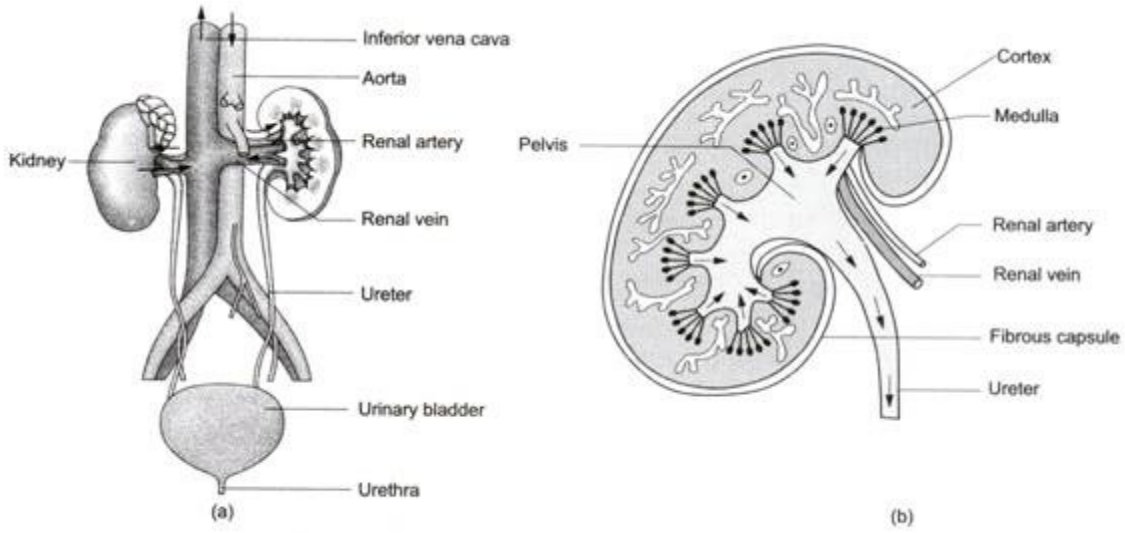
In vertebrates, the main organs of excretion and maintenance of water balance are the kidneys.

Excretion in human beings

Although the kidneys are the main organs of excretion, the skin, lungs and liver also help in excretion.

Skin: Our skin has sweat glands, through which we excrete small amounts of water, urea and salts.
Liver: The liver excretes bile, which contains bile pigments. These are produced by the breakdown of old RBCs in the liver. As hemoglobin breaks down, its iron is retained, while the pigment (haem) is excreted with the bile. The liver also excretes cholesterol.
Lungs: The lungs help in getting rid of carbon dioxide, formed as a result of cellular respiration, through exhalation.
Kidney: Our excretory system consists of kidneys, blood vessels that join them, ureters, urinary bladder and urethra. They help produce and excrete urine.
There are two bean-shaped kidneys that lie in the abdominal cavity, one on either side of the vertebral column. The kidneys are reddish brown. Each of them is about 10 cm long and weighs about 150 g. Although they weigh less, they receive a lot of blood for filtration.

A volume of blood nearly equivalent to that in the whole body passes through the kidneys every four or five minutes. The kidneys produce urine to filter out the waste products, like urea and uric acid, from the blood.



(a) Excretory organs of man (b) Internal structure of a kidney

Urine leaves each kidney through a tube called ureters. The ureters from both the kidneys are corrected to the urinary bladder that collects and stores urine. Ureters carry urine from the kidneys into the urinary bladder. The urethra is a canal that carries urine from the bladder and expels it outside the body.

Excretion in Plants

When plants get rid of harmful and waste substances from their bodies, it's called excretion. Unlike animals, plants don't have a fancy system just for excretion. They don't have special organs dedicated to it. So, excretion in plants isn't as complicated as in animals.

Different Types of Excretion in Plants

Transpiration: Transpiration is when a plant loses water vapor from its surface, mainly from its leaves. This loss of water helps to move water up through the plant's xylem. Plants can lose quite a bit of water through transpiration.

Guttation: Guttation is when a plant loses water in the form of droplets from its hydathodes. Droplets of water come out from the edges of leaves of certain plants. This happens when there's strong pressure in the roots but low transpiration.

Minerals, acids, sugars, and other substances can be found in these droplets. Plants like garden nasturtium, grasses, tomatoes, potatoes, and colocasia can exhibit guttation.

Bleeding: Bleeding is when sap flows out from injured parts of a plant. This happens because of pressure in the roots, causing sap to flow out of cuts made in the stem, especially when the soil is moist.

Respiration: Plants use carbon dioxide, water, sunlight, and chlorophyll to make food in a process called photosynthesis. Oxygen is produced as a result. This oxygen is released as waste through a process called diffusion.

Excretion is carried out in plants in the following ways:

- Plants get rid of their waste gases like oxygen, carbon dioxide, and water vapor through tiny openings called stomata on their leaves and lenticels on their stems.
- Some waste collects in leaves and bark. When leaves and bark fall off, the waste goes away too.
- Other waste turns into harmless solids and stays in the plant. Things like raphides, tannins, resins, gum, rubber, and essential oils are examples of this.
- For instance, the oil from oranges, eucalyptus, and jasmine, latex from rubber trees, papaya trees, and gums from acacia, are all forms of stored waste products. Sometimes, these products even get released into the soil.

Plant Excretory Product

- Plants mainly get rid of carbon dioxide, extra water from breathing, and nitrogen compounds from breaking down proteins. They release two gases: oxygen when they make food and carbon dioxide when they breathe out.
- When animals breathe, they release a gas called carbon dioxide. Plants use this carbon dioxide, along with sunlight, to create their food in a process known as photosynthesis. As a result, plants release oxygen into the air, which is crucial for all living beings.
- Plants absorb water from the soil through their roots. Sometimes, they take in more water than they need. To get rid of this excess water, plants release it through small pores on their leaves called stomata. This process is called transpiration.
- In some cases, the water or soil where plants grow may contain too much salt. When plants absorb this excess salt, they must eliminate it from their systems. They do so by depositing the salts in their tissues, forming crystals.
- As animals and plants go about their daily activities, they produce waste substances. One type of waste is called nitrogenous waste, which includes compounds like urea. These wastes are expelled from the body through processes like diffusion.
- During their metabolic processes, plants produce organic acids. Some of these acids, like oxalic acid, can accumulate in plants and cause harm if not properly managed.
- Plants also produce chemicals called tannins, which act as a defense mechanism against pests. These tannins give certain plants a bitter taste and help protect them from being eaten by animals.
- Certain plants, like rubber trees, produce a milky white fluid called latex. This latex serves to seal wounds in the plant and provides protection against insects and diseases.

- Another protective substance produced by plants is resin, which is a thick liquid with a strong aroma. Resins help shield plants from damage caused by insects and fungi.
- Plants also generate gums, which are sticky substances that help seal wounds and protect against infections. These gums play a crucial role in maintaining the health of the plant.

Plant Organs involved in Excretion

- **Old Leaves:** When leaves get old, they store waste and then fall off the tree.
- **Old Xylem:** Waste like resins and tannins builds up in the old xylem, making it stop working.
- **Bark:** Dead cells in bark collect impurities like tannins, which are used for making dyes and inks.
- **Central Vacuole:** Waste mostly stays in the central vacuole of a plant's cell, but it doesn't affect the cell's work because of a special membrane.
- **Root Excretion:** Plants push some waste out through their roots.
- **Detoxification:** Harmful acids are turned into harmless crystals by plants.
- **Salt Glands:** Plants also get rid of extra salt through special glands.